ELUFISAN, Temidayo Oluyomi, (DSc)

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Personal statement: I am a Doctor of Science in biotechnology who is passionate about research and highly motivated to see how the application of biotechnology can help ensure good health quality and a sustainable living environment. My training as a microbiologist has prepared me with skills such as next-generation genome sequencing technique and analysis, lipid profiling, antimicrobial resistance monitoring, bacteria culturing, identification, and analysis using various molecular techniques such PCR, SDS-PAGE enzyme analysis, and bacteria engineering techniques like cloning and bacteria transformation. I can also use animal models to verify bacteria behavior in vivo.

Education:

- **Doctor of Science in Biotechnology,** Instituto Politecnico Nacional Mexico, **(July 2019),**
- M.**Sc. in Pharmaceutical microbiology,** University of Ibadan, Nigeria, Nigeria **(Oct 2011)**
- **B. Tech, in microbiology,** Ladoke Akintola University of Technology, Nigeria (**Dec 2005)**

Research Experience.

Research fellow at the National Center for Technology Management (NACETEM)
Obafemi Awolowo University, Ile-Ife, Nigeria (**April 2010 – April 2020**)

- Participated in Science Policy writing and was a member of a subgroup which wrote the current Science And innovation policy for Nigeria.
- Participated in human capital development via the training of students for the postgraduate diploma in Technology Management.
- Conduct research on science innovation and technology policy using the social science approach involving the use of questionnaire and data analysis with SPSS software.

ORISE Postdoc fellow with Dr. Jenny Maloney **Feb 2024 - Feb 2026**
United State Department of Agriculture (USDA), Agricultural Research Section (ARS), Beltsville, MD

- Isolate and characterized food borne pathogen of animal and human origin.
- Sequence and analyze parasites' genome.
- Genome assembly, annotation, and comparative analysis of parasites.
- Bioinformatics analysis of parasites.

Postdoc fellow with Dr., Otto Geiger, **Nov 2021 - Oct 2023**
Ecological genomics, Center for Genomic Science, National Autonomous University of Mexico.

- Verified the role of sphingolipid in the virulent behavior of *Caulobacter crescentus* NA 1000 in a larva model using *Galleria mellonella* as a model organism. Larvae were injected with Caulobacter that were deficient in serine palmitoyl transferase and with the wild typed strain to evaluate the effect of sphingolipid on virulence.
- Identified the homologous gene in *Escherichia coli* that may be involved in the biosynthesis of Sphingolipid using an Insilco genome analysis with Synttax and blast search against the NCBI database. Similarly, multiple alignment with MEGA X was used to compare similarities.
- Carried out the extraction of lipopolysaccharides from *Caulobacter crescentus* using the method described by Goldberg and Moore 2012. Other acquired skills in this lab include bacterial lipid profiling and analysis for determining the expression of lipid in bacteria using thin-layer liquid chromatography, and radioactive labelling techniques. Others include RT-PCR for determining the expression of spt gene in *Escherichia coli* BL21 (DE3) SDS-PAGE analysis of lipid and proteins.

Graduate studentship in the lab of Dr Xianwu Guo, Jan 2015 - July 2019
Centro de Biotecnologia Genomica, Instituto Politecnico Nacional, Mexico

- Isolated and characterized *Stenotrophomonas* species from different environmental samples (soil and sewage) using both conventional techniques and molecular techniques using the 16s rRNA amplification and sequencing to confirm the identity of bacteria.
- Evaluated the ability of a *Stenotrophomonas* species to degrade polycylic aromatic hydrocarbon (PAH) in a PAH degrading assay. Metabolites formed from the degradation of PAH were identified by Gas-column chromatography and FTIR machine.
- Evaluated the antimicrobial susceptibility pattern of *Stenotrophomonas* species using the Kirk-Baur agar dilution technique and the microbroth dilution assay.
- Genomic basis for *Stenotrophomonas* ability to degrade PAH and resist several antibiotics was determined by the next-generation sequencing of *Stenotrophomonas* and its analysis.
- Isolated and characterized *Bdellovibrio* species in a prey-pathogen interaction study using the double layer agar plating technique and the bacterial liquid co-culture medium. Mechanism

Graduate Studentship at the Department of Pharmaceutical Microbiology,
University of Ibadan Oct 2008 – Oct 2011

- Isolated and carried out the antibiotic susceptibility profile for Staphylococcus aureus recovered from wounds and HVS swab using conventional technique.

Internship at the Department of Microbiology and Parasitology, University College Hospital, Ibadan, Nigeria 2004-2005

- Isolated and characterized bacteria strain from different human sample using the conventional bacteria plating techniques.

Teaching Experience:

Visiting Scholar at the **Universidad Tecnologico Tamaulipas Norte, Reynosa, México (Oct 2023 – till present**

Visiting Scholar at the **Universidad Tecnologico Tamaulipas Norte, Reynosa, México (Sep. 2019 – December 2021)**

- Trained Undergraduates in English communication and writing.
- Trained students on Seminal presentation and how to attend interviews.

Science Tutor at the **Universidad Tecmilenio Reynosa campus, Reynosa Mexico (August 2017 – Dec 2021)**

- Trained students on mathematical concepts and research methodology
- Exposed students to the introductory concepts of human physiology.

Junior High school Principal at Imperial College Sasa-Ojoo Ibadan, Oyo state (Sep 2007 – March 2010**)**

- Taught the student Biology and prepared them for West African Examination.
- Worked as the Principal of the Junior High overseeing the administrative activity of the Junior High.

Fellowship and Grants.

- National Council for Science and Technology Postdoctoral fellowship 2021 – present
  27,000 Mexican peso per month for 24 months.
- National Council for Science and Technology Doctoral fellowship, Mexico (CONACYT), Mexico 2015 - 2018
  14,000 pesos per month for 4 years.

Honors and Award

- Received the Instituto Politecnico Nacional travel award to attend ASM microbes in June 2018
- Received Travel award from the Federation of European Microbiological Societies to attend the FEMS' conference in 2017.

Supervisory and Mentoring Experience

- Worked with 4 masters' students during their master thesis, helped to put them through on the basic techniques for their study.
- Worked with my Ph.D. supervisor to guide 2 doctorate students during their research work in the lab.

Academic Service

Acted as Secretary (biotechnology Sub-group, Science Policy, and Innovation Studies Department (NACETEM), Obafemi Awolowo University, Ile-Ife, Nigeria,

Served as reviewer for,

- Current microbiology journal (Springer nature)
- CyTA journal of food (Taylor and francis)
- Genes (MDPI)
- Antibiotics (MDPI)
- Bioremediation Journal (Taylor and Francis)
- PLOS ONE

Ongoing research activities

- Whole genome sequence analysis of four multi-drug resistance *Stenotrophomonas* species isolated from Mexico.
- Analysis of the role of sphingolipid in the virulent behavior of the *Caulobacter crescentus* NA 1000
- Identification of the phenotypic and regulatory conditions that can stimulate the biosynthesis of sphingolipid in *Escherichia coli* BL21(DE3)

Certifications and Trainings

Work Ethics and Capacity building organized by Kenokit consult, October 2010

Developing Fundable proposal organized by National Center for Technology Management (An agency of the Federal Ministry of Science and technology, Nigeria) OAU Ile-Ife, February 2011

Ethics and Procedure for Academic Writing and Publishing Organized by the Publication unit of National Center for Technology Management (An agency of the Federal Ministry of Science and technology, Nigeria) OAU Ile-Ife, October 2012

Increasing Productivity through improved worked process organized by Kenokit Consult Nov 2012

Leadership and Activities:

National Youth Service Corps, Osun State (Aug 2006- Aug 2007)

Junior High School Principal, Imperial College Sasa-Ojoo Ibadan (2009-2010)

Bioinformatics and Computer Skills

- Molecular techniques; Possess expertise in Polymerase chain reaction technique, electrophoresis, and DNA sequencing for the characterization of bacteria and bacterial characteristics.
- Possess the ability to use both online software and independent pipelines such as SPAde, Velvet, Prokka for genome assembly, annotation, and analysis.

- I can use python and R for genome analysis at the beginner's level.
- Information Technology – Knowledge of Microsoft Office Packages (Word, Excel SPSS, Power Point.). Regularly used for academic and work projects in various job positions.

Professional affiliation

America Society of Microbiologist. (ASM)
Canadian Society for Microbiologist (CSM)

Presentations

Oral Presentation: Biotechnology A tool for sustainable Algae-based Biodiesel Production in Nigeria; Nigeria Solar Energy Society Conference 2013

Oral presentation: Isolation and Genomic Characterization of *Stenotrophomonas* species from Soil and Sewage from Mexico presented at the IX Jornadas Académicas Doctorado en Ciencias en Biotecnología held at Centro Interdisciplinario de investigacion para el desarrollo integral Regional Unidad Sinaloa May 18-19, 2015.

Oral Presentation: First reports on the Isolation of Bdellovibrio species from soil in Mexico presented at the70th annual meeting of the Subtropical Agriculture and Environments society held at Texas A&M University, Westlaco, Texas United State of America. February 2016

Oral presentation: Advances on the Isolation and Genomic Characterization of *Stenotrophomonas species* from Soil and Sewage from Mexico presented at XI Jornadas Académicas Doctorado en Ciencias en Biotecnología held at Best Western Hotel Plaza Durango between May 16-19, 2016.

Poster Presentation: Isolation and Characterization of *Stenotrophomonas* species from Mexico presented at the 1st Binational conference on Biotechnology held at Centre for Genomics Biotechnology Reynosa, Mexico June 2016

Poster Presentation: Hydrocarbon tolerance test and biodegradation potential of selected strains of *Stenotrophomonas maltophilia* isolated from crude oil contaminated soils from Mexico; accepted for presentation at the 7th Federation of European Microbiologist
Conference, Valencia Spain which took place between 9 and 13 July 2017.

Conference Proceeding: The role of predatory bacteria towards the control of microbial resistance, World Congress and Exhibition on Antibiotics September 14-16, 2015, Las Vegas, USA

Poster Presentation; Vaccine developments against multi-resistant Drug organisms; the role of bacteriophages presented at 9th global expo on Vaccine and Vaccination held at Double Tree Hotel San Franscisco California USA between Nov 30 and 2nd December 2015

Poster Presentation: Characterization and Antibiotic Susceptibility Pattern of Community- isolated *Stenotrophomonas* species; Presented at The University of Texas Rio Grande Valley (UTRGV) School of Medicine's Inaugural Research Symposium held on August 12, 2017

Publications

Elufisan TO, Oyelade, B and Oloke, JK, the protective effective of beta glucan on Escherichia coli induced infection (2010), Journal of Bactriology Research VOL. 3(2), PP.28-31, February 2011

Elufisan, TO, Oyelade, B Oyedara OO (2012) Updates on microbial resistance to Drugs, African Journal of Microbiology research Vol 6(23) pp 4833-4844

Elufisan TO, and Adeleke EO, Susceptibility pattern of Staphylococcus aureus isolated from wound of patient attending treatment at the university college hospital Ibadan Oyo state. African Journal of Medical Science, Vol 4(2) pg 101-115

Elufisan TO & Awoleye, OM, (2012): Combating Cybercrime in a cashless economy- the role of biometrics available at Accounting Technology & Information Systems eJournal Vol 4, Issue 8, April 16, 2012

Awoleye MO, Ogunkambi, DA, Ayo-lawal, RO, Ololade GO, and ***Elufisan TO***, (2012): Techno-Microbial Hazard of Computer Keyboard; Health risk and Policy Implication: American Journal of Information science vol 2 No 4

Elufisan TO and Adeleke EA, 2012: The Epidemiological role of Plasmid curing in the identification of Staphylococcus aureus route of infection. JMEAST vol1 pg 27-44

Oyedara OO, ***Elufisan TO***, Abioye OE, Fadare, TO and Adeyemi FM, 2012: Phytochemical and antibacterial activities of methanolic leaf extract of Lantana camara Linn INTASCI Journal of Microbiology and Biotechnology vol 2(5) pp 1-11

Ayo-Lawal, RA, ***Elufisan TO,*** Ogundari, I, Adesiyan, IM and Famurewa, AJ, 2013: Biotechnology A tool for sustainable Algae-based Biodiesel Production in Nigeria. Nigeria Solar Energy Journal vol 23 pg 6-16

Oluyide BO, Oloke JK, ***Elufisan TO***, Ogunyemi, TF, 2013: Investigation on the Ability of Sulfonolipid from Salinibacter ruber to Ameliorate the Toxic Effect of Sodium Arsenite on the Sperm Count and Testoterone Level in Rat Annual Review & Research in Biology 3 (4), 860-872

Oyedara OO, Luna-Santillana D, de Jesus E, Olguin-Rodriguez O, Guo X, Mendoza-Villa MA, Menchaca-Arredondo JL, ***Elufisan TO***, Garza-Hernandez JA, Garcia Leon I, Rodriguez-Perez MA, 2016. Isolation of Bdellovibrio sp. from soil samples in Mexico and their potential applications in control of pathogens. MicrobiologyOpen. Dec 1;5(6):992-1002.

Sanchez-Varela, A., Rodriguez-Luna, I. C., Ortega-Balleza, J. L., ***Elufisan, TO.,*** Guo, X., Castro- Escarpulli-G. Identificación de Aeromonas spp. A partir de aislados clínicos y ambientales, en Reynosa Tamaulipas, México. Revista de Aplicación Científica y Técnica. 2017, 3-8: 35-39

Omotayo Opemipo Oyedara, Aldo Segura-Cabrera, Xianwu Guo, ***Temidayo Oluyomi Elufisan***, Rafael Alejandro Cantú González, and Mario A. Rodríguez Pérez, "Whole-Genome Sequencing and Comparative Genome Analysis Provided Insight into the Predatory Features and Genetic Diversity of Two BdellovibrioSpecies Isolated from Soil," International Journal of Genomics, vol. 2018, Article ID 9402073, 10 pages, 2018. doi:10.1155/2018/9402073

Elufisan TO, Lozano L, Bustos P, Rodríguez-Luna IC, Sánchez-Varela A, Oyedara OO[4], Villalobos-López MÁ, Guo X. Complete Genome Sequence of *S. maltophilia* Strain SVIA2, Isolated from Crude Oil-Contaminated Soil in Tabasco, Mexico, Microbiol Resour Announc. 2019 Jul 25;8(30). pii: e00529-19. doi: 10.1128/MRA.00529-19.

Elufisan TO, Rodríguez-Luna IC, Oyedara OO, Sánchez-Varela A, Hernández-Mendoza A, DantánGonzalez E, Paz-González AD, Muhammad K, Rivera G, Villalobos-Lopez MA, Guo X. The Polycyclic Aromatic Hydrocarbon (PAH) degradation activities and genome analysis of a novel strain *Stenotrophomonas* sp. Pemsol isolated from Mexico.PeerJ8:e8102 Jan, 2020, http://doi.org/10.7717/peerj.8102.

Vilchis-Carmona, J. A., Rodríguez-Luna, I. C., ***Elufisan, T. O***., Sánchez-Varela, A., Bibbins-Martínez, M., Rivera, G., ... & Guo, X. (2021). The decolorization and degradation of azo dyes by two *Stenotrophomonas* strains isolated from textile effluent (Tepetitla, Mexico). Brazilian Journal of Microbiology, 1-13.

Huerta-alemán, M. E., Martínez-Herrera, R. E., ***Elufisan, T. O***., Gandarilla-Pacheco, F. L., Quintero-Zapata, I., Reyez-López, M. Á., & de Luna-Santillana, E. D. J. (2022). Draft Genome Sequence of a Polyhydroxyalkanoate-Producing Bacillus cereus Strain Isolated from Nuevo Leon State, Mexico. Microbiology Resource Announcements, e00269-22.

Lopez, E. S., ***Elufisan, T. O.***, Bustos, P., Charles, C. P. M., Mendoza-Herrera, A., & Guo, X. (2022). Complete Genome Report of a Hydrocarbon-Degrading Sphingobium yanoikuyae S72. Applied Sciences, 12(12), 6201.

Ajao, Y. O., Rodríguez-Luna, I. C., Elufisan, T. O., Sánchez-Varela, A., Cortés-Espinosa, D. V., Camilli, A., & Guo, X. (2022). Bdellovibrio reynosensis sp. nov., from a Mexico soil sample. International Journal of Systematic and Evolutionary Microbiology, 72(12), 005608.
Martínez-Herrera, R. E., Aemán-Huerta, M. E., Rutiaga-Quiñones, O. M., de Luna-Santillana, E. J., & **Elufisan, T. O**. (2023). A comprehensive view of Bacillus cereus as a polyhydroxyalkanoate (PHA) producer: A promising alternative to Petroplastics. *Process Biochemistry*.

Oyewole, K. A., Oyedara, O. O., Awojide, S. H., Olawade, M. O., Abioye, O. E., Adeyemi, F. M., ... & **Elufisan, T. O.** (2023). Antibacterial and In Silico Evaluation of β-lactamase Inhibitory Potential of Pinus sylvestris L. (Scots Pine) Essential Oil. *Proceedings of the National Academy of Sciences, India Section B: Biological Sciences*, 1-11.

Olea-Ozuna, R. J., Poggio, S., Bergström, E., Osorio, A., ELUFISAN, T. O., Padilla-Gómez, J., ... & Geiger, O. (2024). Genes required for the formation of virulence-provoking bacterial sphingolipids. *bioRxiv*, 2024-02.

Articles under review

nature)

Book chapters

Elufisan, T. O., Oyedara, O. O., & Oluyide, O. O. (2023). 3 Transcriptional Regulation of Phyto-Oxylipins Signaling Pathway. Phyto-Oxylipins: Metabolism, Physiological Roles, and Profiling Techniques.
Oyedara, O. O., Adeyemi, F. M., Adetunji, C. O., & **Elufisan, T. O.** (2022). Repositioning antiviral drugs as a rapid and cost-effective approach to discover treatment against SARS-CoV-2 infection. In Medical Biotechnology, Biopharmaceutics, Forensic Science and Bioinformatics (pp. 155-166). CRC Press.

References

1. Dr. Xianwu Guo, (Doctoral study, Supervisor)
   Center for Genomic Science, National Polytechnic Inistitute, Mexico,
   Email: xguo@ipn.mx

2. Dr Abiodun Egbetokun (Former HOD of my department NACETEM)
   National Center for Technology Management An agency of the Federal Republic of Science and Technology, Nigeria,
   Email: aaegbetokun@gmail.com

3. Dr. Miguel-Angel Villabos-Lopez (Co-Supervisor, Doctoral studies),
   Center for Applied Biotechnology, National Polytechnic Institute, Mexico,
   Email: mangel@ipn.mx